

The Angle Register

Deriving the Base, Shells, and Weights of the BBP Formula from the Smallest Frame Containing a Diagonal

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Abstract

The Bailey-Borwein-Plouffe formula is normally presented as a discovered integer relation: PSLQ found it, verification confirmed it, and the specific values 8, 16, and the coefficient vector $\{4, -2, -1, -1\}$ are treated as facts about the formula rather than consequences of anything. This paper derives all of them from a single geometric requirement and shows that none is a free choice.

The argument runs backward from the target. π is four times $\arctan(1)$, which is four times the angle of the 45-degree direction. A frame of n equally spaced directions contains that direction only when 8 divides n . The integration limit is $\cos(45 \text{ degrees}) = 2^{-1/2}$, and it must be the coordinate of one of those marks. Combined with the requirement that the limit raised to the shell spacing equal the reciprocal of the base, this forces the pairing $\text{base} = 2^m$ with spacing $n = 2m$, and then forces $m = 4$ and $\text{base} = 16$ as the minimum. The denominators $8k+r$ are then one per marked direction, and the shell index satisfies $r = p+1$ for numerator power p purely because integrating x^p produces $x^{(p+1)}/(p+1)$.

The four weights are then computed rather than quoted: $w_r = a_p \text{ times } 2^{(-r/2)}$, where a_p is the numerator coefficient and $2^{(-r/2)}$ is the upper limit raised to the shell index. This reproduces $\{4, -2, -1, -1\}$ exactly, and the resulting sum agrees with π to 25 decimal places. The square roots of two in the numerator polynomial are shown to sit on exactly the odd shells, because odd r makes $2^{(-r/2)}$ irrational and the coefficient must cancel it; and odd r is exactly the set of diagonal directions.

Two further consequences follow. The shells split two and two: odd shells are coprime to 16 and retain full multiplicative orbits, even shells share a factor and fall into trapped cycles. This arithmetic split is identical to the geometric split between diagonals and axes, with parity as

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the bridge. And the sum cannot terminate, because 16 is never a primitive root modulo any prime of the form $8k+1$ (2 is a quadratic residue there), so every shell is permanently blind to most of its own residues and no finite collection is complete.

The conclusion is that the integers appearing in BBP are not magnitudes. The decomposition $n = 8k + r$ is polar: the quotient k records completed cycles and is projected out by the eighth root of unity, and the residue r selects an orientation. Comparative searches over base 4 with 4 sectors and base 8 with 6 sectors return only logarithms; only the base 16 frame, the first one containing a diagonal, returns pi.

§1 The Question

The Bailey-Borwein-Plouffe formula (Bailey, Borwein and Plouffe, 1997) states:

$$\pi = \text{SUM over } k \text{ of } (1/16^k) [4/(8k+1) - 2/(8k+4) - 1/(8k+5) - 1/(8k+6)]$$

It permits extraction of the n -th hexadecimal digit of π without computing the preceding digits, and it was found by an integer relation algorithm rather than derived from an understanding of what π is. The standard account of its origin is a substitution and a partial fraction expansion. That account is correct and it is not an explanation. It does not say why the denominators advance by 8, why the base is 16, why exactly four of the eight residue classes survive, or why the surviving weights are 4, -2, -1, -1 and not some other integers.

This paper takes the position that these are not facts about the formula but consequences of a single choice made at the start, and traces each one back to that choice. The method throughout is to state the requirement, derive the consequence, and verify numerically. Where a claim is measured rather than proved, it is marked as such.

Notation. Throughout, u denotes the upper integration limit $2^{(-1/2)}$; n denotes the number of equally spaced directions in the frame (equivalently the shell spacing); m denotes the base exponent, so that the base b equals 2^m ; r denotes the residue class modulo n ; k denotes the shell index; and p denotes a power in the numerator polynomial. Angles are given in degrees.

§2 The Polar Decomposition of the Shell Index

Every integer admits a unique decomposition $n = 8k + r$ with r in $\{0, \dots, 7\}$. This is elementary. What is not elementary is what happens to that decomposition inside the formula.

Let ω denote the primitive eighth root of unity, $\omega = \exp(2\pi i/8)$, a rotation by 45 degrees. Then:

$$\omega^{(8k+r)} = (\omega^8)^k * \omega^r = 1^k * \omega^r = \omega^r$$

The quotient k contributes nothing. Complete turns cancel. Only the residue r survives, and it survives as an angle: r times 45 degrees.

This was verified directly across the shell family. For $r = 1$ and $k = 0$ through 6, the shell indices are 1, 9, 17, 25, 33, 41, 49 -- seven distinct magnitudes -- and the corresponding angles are:

$8k+r$	magnitude	angle of $\omega^{(8k+r)}$
1	1	45.00 degrees

$8k+r$	magnitude	angle of $\omega^{(8k+r)}$
9	9	45.00 degrees
17	17	45.00 degrees
25	25	45.00 degrees
33	33	45.00 degrees
41	41	45.00 degrees
49	49	45.00 degrees

Seven different magnitudes, one angle. The decomposition $n = 8k + r$ is polar: k is the winding count and r is the orientation within the current turn. The shell index is a coordinate, not a quantity.

Precision on wording. It is tempting to say that the magnitude is annihilated. It is not. The map $8k + r$ maps to r is a quotient map, and k remains fully recoverable from the integer; it is simply not used by the operation. The distinction matters: the information is projected out by a choice of representation, not destroyed by the mathematics. This is the same distinction that separates a discarded chaining value from an information-theoretic loss.

Precision on the operator. It is also tempting to say that a direction rotates. Directions do not rotate; operators do. The correct statement is that the rotation operator is applied repeatedly, and the diagonal direction is invariant under the eighth power of that operator. The integers index applications of the operator. They carry operator depth, not quantity.

§3 The Base and the Spacing Are One Choice

The formula uses the geometric factor $1/16^k$ and the denominator family $8k+r$. These appear as two independent parameters. They are not independent.

The series arises from the expansion $1/(1 - x^8) = \text{SUM over } k \text{ of } x^{(8k)}$, evaluated on the interval from 0 to u . At the upper limit the geometric factor is u^8 . For the formula to produce base-16 digits, this must equal $1/16$:

$$u^8 = (2^{(-1/2)})^8 = 2^{(-4)} = 1/16$$

Verified: $u^8 = 0.0625000000$ exactly. Generalising, with $u = 2^{(-1/2)}$ and base $b = 2^m$, the requirement $u^n = 1/b$ gives $2^{(-n/2)} = 2^{(-m)}$, hence:

$$n = 2m$$

The base exponent and the shell spacing are locked. Verified across the family:

base b	m	spacing n = 2m	u^n	equals 1/b
2	1	2	0.5000000000	yes
4	2	4	0.2500000000	yes
8	3	6	0.1250000000	yes
16	4	8	0.0625000000	yes
32	5	10	0.0312500000	yes
64	6	12	0.0156250000	yes

Note that base 8, being 2^3 , pairs with spacing 6 and not spacing 8. The rule is base 2^m pairs with $2m$, not that a base pairs with itself. This distinction is easy to lose and it matters for what follows.

§4 The Diagonal Requirement and the Minimality of Base 16

This section contains the central derivation. Two independent conditions must be satisfied simultaneously, and together they select $n = 8$ uniquely as the minimum.

§4.1 Pi is an angle, and it is the diagonal angle

The identity $\pi = 4 * \arctan(1)$ is elementary. Numerically, $4 * \arctan(1) = 3.141592653589793238462643$, matching π to 25 digits. What matters is its content: $\arctan(1)$ is the angle whose tangent is 1, which is the angle of the direction (1,1) -- the diagonal, at 45 degrees. π is four times the diagonal angle.

A frame consisting of n equally spaced directions has marks at multiples of $360/n$ degrees. It contains the 45-degree direction if and only if $360/n$ divides 45, equivalently if and only if 8 divides n .

sectors n	step	marks (degrees)	contains 45?
2	180	0, 180	no
4	90	0, 90, 180, 270	no

sectors n	step	marks (degrees)	contains 45?
6	60	0, 60, 120, 180, 240, 300	no
8	45	0, 45, 90, 135, 180, 225, 270, 315	YES
10	36	0, 36, 72, 108, 144, ...	no
12	30	0, 30, 60, 90, 120, 150, ...	no

Eight is the smallest sector count containing a 45-degree direction.

§4.2 The limit must land on the frame

The second condition is independent of the first and reaches the same place. The upper limit $u = 2^{(-1/2)} = 0.7071067812$ is the x-coordinate of the 45-degree direction: $\cos(45 \text{ degrees}) = 2^{(-1/2)}$. For the integration to terminate on a mark of the frame rather than at an arbitrary interior point, u must appear among the mark coordinates $\cos(j \cdot 360/n)$.

sectors n	mark x-coordinates	$u = 0.7071$ present?
2	-1, 1	no
4	-1, 0, 1	no
6	-1, -0.5, 0.5, 1	no
8	-1, -0.707107, 0, 0.707107, 1	YES

At $n = 4$ the available coordinates are -1, 0 and 1; the value 0.7071 is not among them. At $n = 6$ they are -1, -0.5, 0.5 and 1; again absent. At $n = 8$ the coordinates plus and minus 0.707107 appear for the first time.

§4.3 The forced conclusion

Both conditions require 8 to divide n . Combined with the lock $n = 2m$ from Section 3, this forces m to be a multiple of 4, hence the base must be 16, 256, 4096, and so on. The minimum is:

$$n = 8, \quad m = 4, \quad \text{base} = 16$$

Base 16 is the smallest base in which 45 degrees exists as a direction. This is the answer to the question of why the formula uses hexadecimal. It is not a convenience for binary

computers, and it is not an artefact of the particular substitution used in the derivation. It is the minimum frame in which the target angle has a name.

§5 Comparative Search: What Each Frame Produces

The preceding argument predicts that frames without a diagonal cannot reach pi. This is testable. Since the series is linear in its coefficient vector, the basis constants $B_r = \sum \text{over } k \text{ of } 1/(b^k * (n*k + r))$ can be precomputed once per frame, after which searching a coefficient box is a fast linear scan.

Three frames were searched with an integer coefficient range from -8 to 8, against a target list containing pi and its small multiples, log 2, log 3 and small combinations of these.

frame	vectors searched	constants found	pi found?
base 4, 4 sectors	28,561	$\ln 3, 2 \ln 2$	no
base 8, 6 sectors	4,826,809	$4 \ln 2$	no
base 16, 8 sectors	5,764,801	$2 \ln 3, \text{ and } \pi$	YES

The base 16 hit was the vector $\{4, 0, 0, -2, -1, -1, 0, 0\}$, which is the BBP coefficient set, together with its doubling giving $2*\pi$. Within the base 16 frame the two hits separate by geometry: the $2 \ln 3$ relation uses residues 2 and 6, which are the 90 and 270 degree directions -- pure axes -- while the pi relation uses residues 1 and 5, the 45 and 225 degree diagonals, together with two axis terms. Pi is the only hit that touches a diagonal.

Scope. These are searches over bounded coefficient boxes, not proofs. The correct statement is that no relation for pi was found at base 4 or base 8 within the ranges searched. Whether the absence is a theorem or a limitation of the search box is open; see Section 11.

§6 The Weights, Derived

The coefficient vector $\{4, -2, -1, -1\}$ is normally quoted. It can be computed. The derivation also produces, as a by-product, the relation between the numerator support and the surviving shell indices.

Consider a single monomial integrated over the shell denominator:

$$\text{INTEGRAL from } 0 \text{ to } u \text{ of } x^p / (1 - x^8) dx$$

Expanding the denominator as a geometric series and integrating term by term:

$$\begin{aligned}
 &= \text{SUM over } k \text{ of } u^{(8k+p+1)} / (8k + p + 1) \\
 &= u^{(p+1)} * \text{SUM over } k \text{ of } (u^8)^k / (8k + (p+1)) \\
 &= 2^{(-(p+1)/2)} * \text{SUM over } k \text{ of } (1/16)^k / (8k + (p+1))
 \end{aligned}$$

Two results fall out of this single line.

Result 1: $r = p + 1$. The shell index equals the numerator power plus one. The reason is that integrating x^p produces $x^{(p+1)}/(p+1)$. The offset by one is the integration and nothing else. This explains the correspondence noted between the numerator support $\{0, 3, 4, 5\}$ and the surviving shells $\{1, 4, 5, 6\}$: it is exact for all four terms, with no exceptions, because it is mechanical.

Result 2: the weight formula. Each shell arrives multiplied by $u^{(p+1)} = 2^{(-r/2)}$. Therefore the weight on shell r is:

$$w_r = a_p * 2^{(-r/2)}$$

where a_p is the coefficient of x^p in the numerator polynomial. Applying this to the standard numerator $N(x) = 4*\sqrt{2} - 8x^3 - 4*\sqrt{2}*x^4 - 8x^5$:

power p	numerator coeff a_p	shell r = p+1	limit factor $2^{(-r/2)}$	weight w_r
0	$4*\sqrt{2} = 5.65685425$	1	0.707106781	4
3	-8	4	0.250000000	-2
4	$-4*\sqrt{2} = -5.65685425$	5	0.176776695	-1
5	-8	6	0.125000000	-1

The recovered weight vector is $\{4, -2, -1, -1\}$, matching the BBP coefficients exactly. Summing the weighted basis constants gives $3.141592653589793238462643$ against $\pi = 3.141592653589793238462643$, agreeing to 25 decimal places.

The weights are therefore not free parameters and not the output of a search. They are numerator coefficients scaled by powers of the integration limit, and the limit is fixed by the geometry of Section 4.

§7 Why the Numerator Carries Square Roots of Two

The numerator polynomial contains $\sqrt{2}$ in two of its four terms and plain integers in the other two. This is not decoration; it is forced, and its placement is diagnostic.

The weight formula $w_r = a_p * 2^{(-r/2)}$ contains the factor $2^{(-r/2)}$. For even r this factor is rational: $2^{(-2)} = 1/4$ and $2^{(-3)} = 1/8$. For odd r it is irrational: $2^{(-1/2)}$ and $2^{(-5/2)}$. Since the weights themselves are integers, the numerator coefficient must supply exactly the irrational factor needed to cancel.

shell r	parity	$2^{(-r/2)}$	rational?	numerator coeff	carries sqrt2?
1	odd	0.707106781	no	$4*\sqrt{2}$	YES
4	even	0.250000000	yes	-8	no
5	odd	0.176776695	no	$-4*\sqrt{2}$	YES
6	even	0.125000000	yes	-8	no

The correspondence is perfect: $\sqrt{2}$ appears in the numerator at exactly the odd shells and nowhere else. And by Section 2, odd r corresponds to the 45-degree-offset directions -- the diagonals.

So $\sqrt{2}$ is the diagonal signature. It is $1/\cos(45 \text{ degrees})$, the reciprocal of the diagonal coordinate, and it appears in the algebra at precisely the terms that sit on diagonal directions. The polynomial carries the geometry.

§8 Rotors and Dampers: Parity as the Bridge

The four surviving shells split two and two, and the split can be read either arithmetically or geometrically. They are the same split.

Arithmetically: shells with odd r produce odd moduli $8k+r$, which are coprime to 16 for every k . The multiplicative order of 16 modulo such a modulus is defined, and the orbit of 16 returns to 1. Shells with even r produce even moduli, which share a factor with 16 (specifically 4 for $r = 4$ and 2 for $r = 6$). Sixteen is not invertible modulo such a modulus, the orbit never returns to 1, and the sequence falls into a short trapped cycle from which it cannot escape.

Geometrically: odd r corresponds to angles that are odd multiples of 45 degrees -- the diagonals. Even r corresponds to angles that are multiples of 90 degrees -- the axes.

r	angle	geometry	$8k+r$ parity	gcd with 16	orbit behaviour	weight
1	45	DIAGONAL	odd	1	returns to 1	+4
5	225	DIAGONAL	odd	1	returns to 1	-1
4	180	AXIS	even	4	trapped cycle	-2
6	270	AXIS	even	2	trapped cycle	-1

Parity is the bridge. Being at an odd multiple of 45 degrees and being coprime to 16 are the same condition on r , and they produce the same consequence: a shell that retains its full multiplicative structure and continues to move under repeated application, versus one that loses invertibility and settles into a short repeat.

A further structural note: residues 1 and 5 are 180 degrees apart. They are not two independent directions but one line -- the diagonal through (1,1) and (-1,-1) -- read from both ends, weighted +4 and -1. Residues 4 and 6 are the negative x and negative y axes. The formula is therefore one diagonal read from both ends plus two axes, not four unrelated terms.

The upper limit $u = \cos(45 \text{ degrees})$ is the coordinate of the diagonal direction. The integration proceeds along the diagonal; the axis terms are what it is subtracted against.

§9 Why Pi and Not a Logarithm

The comparative search in Section 5 returned logarithms from the frames without a diagonal and pi only from the frame with one. The mechanism is visible in the partial fraction expansion.

Writing $N(x)/(1 - x^8)$ as a sum of terms $R_j/(x - \omega_j)$ over the eighth roots of unity, each term integrates to $R_j * \log(x - \omega_j)$. For a complex argument:

$$\log(z) = \log|z| + i * \arg(z)$$

The real part is a logarithm of a magnitude -- a length. The imaginary part is an argument -- an angle. A pole lying on the real axis has no argument to contribute and can only ever produce a length. A pole off the real axis contributes an angle.

Since pi is four times the 45-degree angle, reaching it requires a term whose argument is 45 degrees, which requires a pole at 45 degrees, which requires the frame to contain that

direction. A frame without a diagonal can still measure lengths along its axes -- and does, producing $\log 2$ and $\log 3$ -- but it has no name for an angle that is not one of its marks.

This is consistent with the search results: base 4 and base 8 returned only logarithms; base 16 returned both a logarithm relation (using only axis residues) and π (using diagonal residues).

§10 Why the Sum Cannot Terminate

A separate question is why the series is infinite rather than closing after finitely many shells. The answer is a fact about the step size.

Proposition. *Sixteen is never a primitive root modulo a prime of the form $8k+1$.*

Proof. Let p be prime with p congruent to 1 modulo 8. By the second supplement to quadratic reciprocity, 2 is a quadratic residue modulo p , hence the multiplicative order of 2 divides $(p-1)/2$. Since $16 = 2^4$, the order of 16 divides the order of 2, hence divides $(p-1)/2$. Therefore the orbit of 16 covers at most half of the $(p-1)$ available residues, and 16 cannot be a primitive root.

Verified computationally: of 161 primes of the form $8k+1$ below 5000, zero have 16 as a primitive root. The maximum observed orbit coverage was $1/8$, well below the theoretical bound of $1/2$, with several shells at $1/40$ or lower.

The consequence is structural. If 16 were a primitive root modulo some shell, that shell would reach every residue and could in principle carry the entire readout by itself, permitting a closed form. Since it never is, every shell is permanently blind to most of its own residue space, and no finite collection of shells is guaranteed complete. Positional digit extraction in base 16 and a terminating formula are mutually exclusive.

Since base 16 is forced by the diagonal requirement of Section 4, and the step being a power of two is what makes it non-primitive, the infinity of the series traces back to the same geometric choice as everything else.

§11 Epistemic Status

Claim	Status	Basis
$8k+r$ is polar: k is winding, r is orientation	[V]	$\omega^{(8k)} = 1$; 7 magnitudes, one angle
base 2^m pairs with spacing $2m$	[V]	$u^n = 1/b$, verified $m = 1..6$

Claim	Status	Basis
45 degrees is a mark only when 8 divides n	[V]	360/n divides 45
$u = 2^{(-1/2)}$ is a mark coordinate only when 8 divides n	[V]	cos tables, $n = 2, 4, 6, 8$
Base 16 is the minimum base with a diagonal	[V]	both conditions force $8 \mid n$; $n = 2m$
$r = p + 1$ from integration	[V]	integral of x^p gives $x^{(p+1)}/(p+1)$
$w_r = a_p * 2^{(-r/2)}$, reproducing $\{4, -2, -1, -1\}$	[V]	computed; sum matches pi to 25 digits
sqrt2 sits on exactly the odd (diagonal) shells	[V]	$2^{(-r/2)}$ irrational iff r odd
Odd shells rotate, even shells trap; parity is the bridge	[V]	gcd(16, 8k+r) and orbit computation
16 is never a primitive root mod $8k+1$	[V]	reciprocity; 161 primes, zero exceptions
Base 4 and base 8 return only logarithms	[M]	bounded search, not a proof
The BBP coefficient vector is minimal	[M]	2 hits in 28,561; not unique over R
Angles require off-axis poles; lengths do not	[I]	partial fractions; interpretive

[V] verified by computation or proof. [M] measured within a stated bound. [I] interpretive reading of a standard result.

On the uniqueness of the numerator. The constraint is a single equation, SUM over p of $a_p * 2^{(-(p+1)/2)} * B_{(p+1)} = \pi$, in eight unknowns. Over the reals it has infinitely many solutions, so the numerator polynomial is not unique in any absolute sense. What was established is that within the small-integer box searched, exactly two vectors appear -- the BBP set and its doubling. The relation is minimal, not unique, and the paper should not be read as claiming otherwise.

§12 Open Problems

Omega 1 (Absence at lower bases). No relation for π was found at base 4 or base 8 within the coefficient ranges searched. The geometric argument of Section 4 suggests this should be a theorem: without a 45-degree mark, no term can contribute the required argument.

Converting that suggestion into a proof requires showing that no admissible numerator over the smaller frame can produce a term with argument 45 degrees, which has not been done here.

Omega 2 (Classification by frame). The comparative search returned logarithms from axis-only frames and π from the frame containing a diagonal. This suggests a general classification: which constants are reachable from which sector counts, indexed by the angles their defining arctangents require. A frame of n sectors names the angles that are multiples of $360/n$; the conjecture is that a BBP-type formula for a constant exists in that frame only if the constant is expressible using those angles. This is stated, not proved.

Omega 3 (Higher frames). The condition 8 divides n admits $n = 16, 24, 32$ and so on, corresponding to bases 256, 4096 and beyond. These frames contain the diagonal and should therefore admit π relations. Whether they do, and whether their coefficient vectors bear any systematic relation to the base 16 vector, has not been examined.

Omega 4 (The projection and its inverse). The map $8k + r$ maps to r discards the winding count by projection. Section 2 notes that k remains recoverable from the integer. Whether any digit-extraction scheme makes use of the winding count -- that is, whether there is a formula sensitive to k and not only to r -- is unexplored, and would represent a different class of construction than the one considered here.

§13 Conclusion

Every numerical parameter in the BBP formula traces back to one requirement, and the chain has no free choices in it:

π is four times the 45-degree angle. A frame contains that direction only when 8 divides its sector count. The integration limit is the cosine of that angle, and it must lie on the frame, which imposes the same condition. The limit raised to the sector count must equal the reciprocal of the base, locking base 2^m to spacing $2m$. Together these force spacing 8, exponent 4, and base 16 as the minimum. The denominators $8k+r$ are one per marked direction. The shell index exceeds the numerator power by one because that is what integration does. The weights are numerator coefficients scaled by the limit raised to the shell index, giving $\{4, -2, -1, -1\}$ exactly. The square roots of two appear at the odd shells because

odd shells carry irrational limit factors, and odd shells are the diagonals. Odd shells stay coprime to the step and keep turning; even shells lose invertibility and trap. And the series cannot terminate because a power of two is never a primitive root in the relevant moduli, so no shell ever sees its whole residue space.

The integers appearing throughout are not magnitudes. The quotient records completed cycles and is projected away; the residue selects an orientation. They carry traversal state rather than value. The formula is not a computation performed upon numbers -- it is a reading taken from a frame, and the frame is the smallest one in which the angle being measured has a name.

References

- Bailey, D.H., Borwein, P.B., Plouffe, S. (1997). On the Rapid Computation of Various Polylogarithmic Constants. *Mathematics of Computation*, 66(218), 903-913.
- Bailey, D.H., Crandall, R.E. (2001). On the Random Character of Fundamental Constant Expansions. *Experimental Mathematics*, 10(2), 175-190.
- Borwein, J.M., Bailey, D.H. (2004). *Mathematics by Experiment: Plausible Reasoning in the 21st Century*. AK Peters.
- Ferguson, H.R.P., Bailey, D.H., Arno, S. (1999). Analysis of PSLQ, an Integer Relation Finding Algorithm. *Mathematics of Computation*, 68(225), 351-369.
- Ireland, K., Rosen, M. (1990). *A Classical Introduction to Modern Number Theory* (2nd ed.). Springer. [quadratic reciprocity, second supplement]
- Kulik, D. (2026). The Address Decoder: How an Integer Reads a Digit of Pi Without Computing the Preceding Ones. *QuHarmonics-BBP-v1*.
- Kulik, D. (2026). One Edge Operator: Scalar Field Dynamics and Conservation from Edge Memory. *QuHarmonics*.
- Kulik, D. (2026). The Co Audit Trail. *QuHarmonics-Co-AUDIT-v1*.
- Lambert, J.H. (1761). Memoire sur quelques proprietes remarquables des quantites transcendentes circulaires et logarithmiques. *Histoire de l'Academie Royale des Sciences et des Belles-Lettres de Berlin*.